



# Modelling Financial Markets

## Chapter 9: Realised Volatility



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## Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BV: Bipower Variation
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ CLT: Central Limit Theorem
- ▣ DJIA: Dow Jones Industrial Average
- ▣ DM: Diebold–Mariano (test)
- ▣ ETF: Exchange-Traded Fund
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FDR: False Discovery Rate
- ▣ FOMC: Federal Open Market Committee
- ▣ FRED: Federal Reserve Economic Data
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HAR-CJ: HAR model with Continuous and Jump components
- ▣ HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- ▣ HARQ: HAR model with Realised Quarticity
- ▣ HEAVY: High-frEQUENCY-bAsed VolatilitY (models)
- ▣ IQ: Integrated Quarticity
- ▣ LLM: Large Language Model
- ▣ MAE: Mean Absolute Error
- ▣ MCS: Model Confidence Set

## Acronyms used in this material (2/2)

- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ RF: Random Forest
- ▣ RK: Realised Kernel
- ▣ RQ: Realised Quarticity
- ▣ RV: Realised Variance
- ▣ SE: Standard Error
- ▣ SHAR: Semivariance HAR model
- ▣ TRV: Truncated Realised Variance
- ▣ TSRV: Two-Scale Realised Variance
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility IndeX
- ▣ VRP: Variance Risk Premium

## Outline

Motivation

Realised Variance

Microstructure Noise

Jumps

Stylised Facts of Realised Volatility

The HAR-RV Model

Forecasting: HAR vs GARCH

Rough Volatility

Case Study: Machine Learning vs HAR

AI for Scientific Discovery

Summary

References

## Volatility Is Latent: Can We Measure It?

### The problem

- ▣ Volatility drives risk (Chapters 7–8), option prices (Chapter 12) and portfolio weights (Chapter 4)
- ▣ But it is never observed: one daily return is one draw from a distribution whose width we want
- ▣ GARCH (Chapter 5) **infers** today's variance from past daily returns through a model

### The idea of this chapter

- ▣ Inside one day there are hundreds of price changes: add up their squares
- ▣ The sum **measures** the day's variance almost without a model: realised variance [Andersen & Bollerslev \(1998\)](#)
- ▣ Measured volatility can then be modelled and forecast like any time series
- ▣ Early idea: [Merton \(1980\)](#) showed that variance, unlike the mean, is estimated better with more frequent data

## Realised Volatility: From Sceptics to Standard



Torben G. Andersen, 2012

Photo: Soralum (2012); CC BY-SA 4.0; Wikimedia

[Commons](#)

- Measure first, then model: Andersen, Bollerslev, Diebold & Labys (2001), Andersen, Bollerslev, Diebold & Ebens (2001) study the distribution of realised volatility; Andersen, Bollerslev, Diebold & Labys (2003) forecast it directly
- Why the sceptics were wrong (Section 7) and textbook treatment: Franke, Härdle & Hafner (2019)

## Data Used in This Chapter

| Series                   | Frequency                                | Period         | Days               |
|--------------------------|------------------------------------------|----------------|--------------------|
| SPY (ETF on the S&P 500) | 5-minute bars, 09:30–16:00 New York time | 2020–2026      | 1 489              |
| Bitcoin                  | 5-minute bars, 24-hour UTC days          | 2024–2026      | 659                |
| SPY, Bitcoin             | daily closes                             | 2000/2014–2026 | for GARCH          |
| VIX                      | daily close                              | 2020–2026      | implied volatility |

- Up to 18 September 2026; returns  $r_i = 100(\ln P_i - \ln P_{i-1})$  in %; a bar is labelled by its start time
- SPY: 78 returns per full day, 42 on half-days; bars off the 5-minute grid removed; missing bars inside a day carry the last price
- Bitcoin: 288 returns per day; days with fewer than 95% of the bars dropped (88 of 747 calendar days, mostly a data gap in October 2025)

## A Model for Intraday Prices

- Notation: day  $t$  has  $M$  intervals of length  $\Delta = 1/M$ , returns  $r_{t,i}$ , daily return  $r_t = \sum_i r_{t,i}$ 
  - ▶ Variances in %<sup>2</sup>; annualised volatility  $\sqrt{252 \times \text{variance}}$  (SPY),  $\sqrt{365 \times \text{variance}}$  (Bitcoin); the SPY overnight return (close to next open) is outside the bars
- Log price in continuous time:  $dp(s) = \mu(s) ds + \sigma(s) dW(s)$ , with  $W$  a Brownian motion (Chapter 11)
  - ▶  $\sigma(s)$  is the **spot volatility**: it changes within the day and from day to day
- **Integrated variance** of day  $t$ : the quantity we want to measure
  - ▶  $IV_t = \int_{t-1}^t \sigma^2(s) ds$
- If  $\sigma$  is independent of  $W$  (no leverage) and the drift is zero (or fixed by the path of  $\sigma$ ), then given that path the daily return is Normal:  $r_t \mid \sigma \sim N(\int \mu, IV_t)$ 
  - ▶ With leverage (Section 6) this conditional normality fails, but the limit theory of RV below still holds
- $IV_t$  is the natural daily variance; GARCH's  $\sigma_t^2$  is a model-based forecast of it from yesterday's information

## Definition: Realised Variance

- **Realised variance:** the sum of squared intraday returns

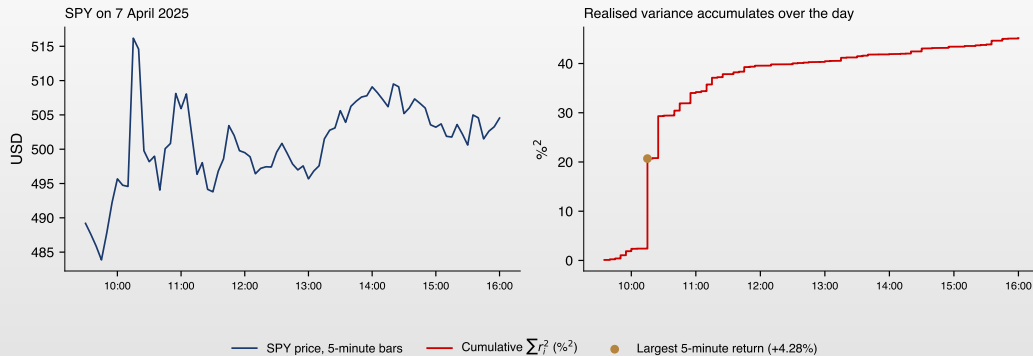
- ▶ 
$$RV_t = \sum_{i=1}^M r_{t,i}^2, \text{ realised volatility} = \sqrt{RV_t}$$

- As  $M \rightarrow \infty$ ,  $RV_t \rightarrow IV_t$ : the drift vanishes and the sum of squares converges to the **quadratic variation** Barndorff-Nielsen & Shephard (2002)

- ▶ The mean of SPY 5-minute returns is below 1% of their standard deviation: it can be ignored

- No parameters to estimate:  $RV_t$  is a **model-free** measurement of one day's variance
- The squared daily return  $r_t^2$  is the special case  $M = 1$ : unbiased (with zero drift) but extremely noisy

## One Day, 78 Returns: 7 April 2025

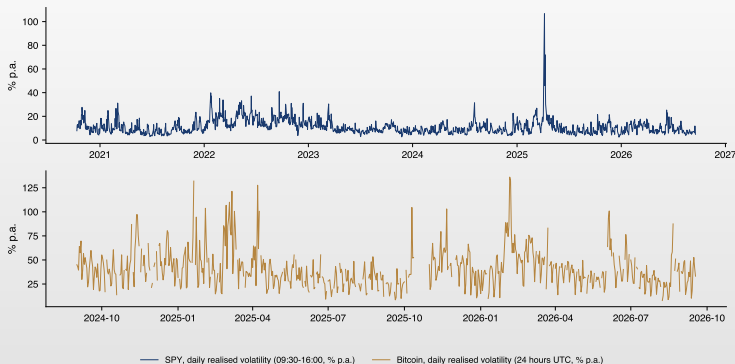


□ Left: SPY on 5-minute bars on the most volatile day of the sample; right: running sum of squared returns

## Reading the Chart: 7 April 2025

- SPY opened  $-3.27\%$  below the previous close, after tariff announcements; 78 five-minute returns followed
  - ▶  $RV = 45.1\%^2$ : annualised realised volatility  $\sqrt{252 \times 45.1} = 107\%$
- One 5-minute return of  $+4.28\%$  (interval ending 10:15) contributed 41% of the day's realised variance
- Open-to-close the day ended  $+3.09\%$ : the daily return alone would call it a normal day, realised variance calls it one of the most turbulent days since 2020

## Daily Realised Volatility: SPY and Bitcoin

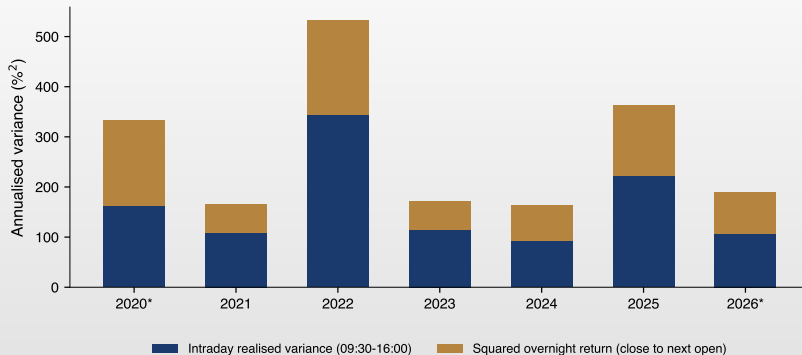


□ Annualised, from 5-minute returns; SPY: regular US session; Bitcoin: 24 hours; the gap in October 2025 is missing data

## Reading the Realised Volatility Series

- SPY: volatility from mean RV  $\sqrt{252 \overline{RV}} = 12.9\%$  p.a., median day 9.7%, range 2.4% to 106.7%
  - ▶ Above the median day: a few very volatile days dominate the mean variance (a root-mean-square, not a mean of daily volatilities)
- Bitcoin:  $\sqrt{365 \overline{RV}} = 45.1\%$  p.a., median 37.8%, maximum 136.1% on 2026-02-05
- Both series show **clustering**, as with GARCH (Chapter 5), but every point is a measurement, not a model output

## Where Is the Risk: Day or Night?



- SPY, mean daily variance by year, annualised: intraday realised variance plus squared overnight return; 2020 and 2026 are partial years

## Reading the Overnight Decomposition

- Overnight returns carry 38% of SPY's daily variance, although the market is closed for 17.5 hours
  - ▶ News (earnings, macro data, Asian and European sessions) accumulates and is priced at the open
- Annualised: close-to-close 16.6%; open-to-close 13.6%; overnight 10.1%
- Largest overnight move:  $-4.08\%$  on 2024-08-05
- For daily risk we use **total** variance:  $RV_t + r_{\text{overnight},t}^2$  (annualised mean 16.4%)
  - ▶ Alternatives: scale intraday RV up, or weight the two parts optimally [Hansen & Lunde \(2005\)](#)

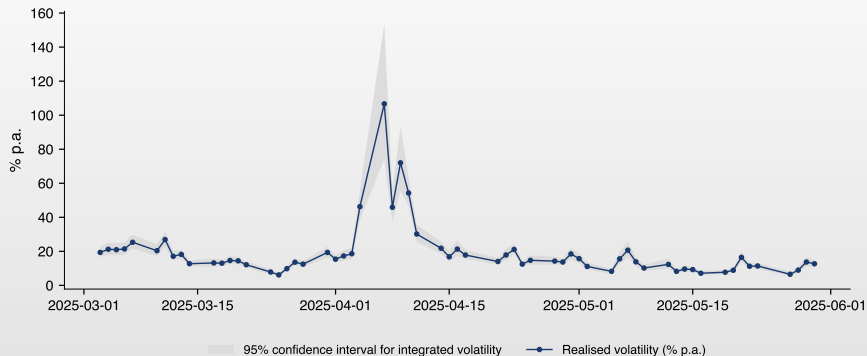
## The Setting: Itô Semimartingales and Stable Convergence

- Itô semimartingale:  $dp = \mu ds + \sigma dW + dJ$ , with  $\mu, \sigma$  adapted and  $\sigma$  càdlàg;  $\sigma$  may depend on  $W$  (leverage)
  - ▶ Without jumps:  $\sqrt{M}(RV_t - IV_t) \xrightarrow{\mathcal{L}-s} MN(0, 2IQ_t)$ , a **mixed** Normal distribution whose variance  $2IQ_t$ , with integrated quarticity  $IQ$ ,  $IQ_t = \int \sigma^4(s) ds$ , is itself random [Barndorff-Nielsen & Shephard \(2002\)](#)
- **Stable** convergence ( $\mathcal{L}$ -s): the limit holds jointly with every variable of the path's  $\sigma$ -field [Mykland & Zhang \(2009\)](#)
  - ▶ Hence  $(RV_t - IV_t)/\sqrt{2RQ_t/M} \rightarrow N(0, 1)$ : studentising by  $\sqrt{2RQ_t/M}$ , with  $RQ_t \rightarrow IQ_t$  random, is legitimate only under stable convergence
- Finite samples: the  $t$ -statistic for  $\ln RV_t$  has better coverage than the one in levels [Barndorff-Nielsen & Shephard \(2005\)](#); we use it on the next slide

## How Precise Is Realised Variance?

- Feasible central limit theorem for RV [Barndorff-Nielsen & Shephard \(2002\)](#):  
 $\sqrt{M}(\text{RV}_t - \text{IV}_t)/\sqrt{2\text{RQ}_t} \rightarrow N(0, 1)$ 
  - ▶  $\text{IQ}_t$  is estimated by realised quarticity  $\text{RQ}_t = \frac{M}{3} \sum_i r_{t,i}^4$
- Standard error:  $\text{se}(\text{RV}_t) = \sqrt{\frac{2}{3} \sum_i r_{t,i}^4}$
- 95% CI on the log scale (always positive):  $\text{RV}_t \exp(\pm 1.96 \text{se}(\text{RV}_t)/\text{RV}_t)$ 
  - ▶ With constant volatility,  $\text{se}/\text{RV} \approx \sqrt{2/M}$ : 16% for  $M = 78$
- RV is a measurement **with error**: it matters when RV is the target of a forecast (Section 6)

## Confidence Intervals for Daily Volatility

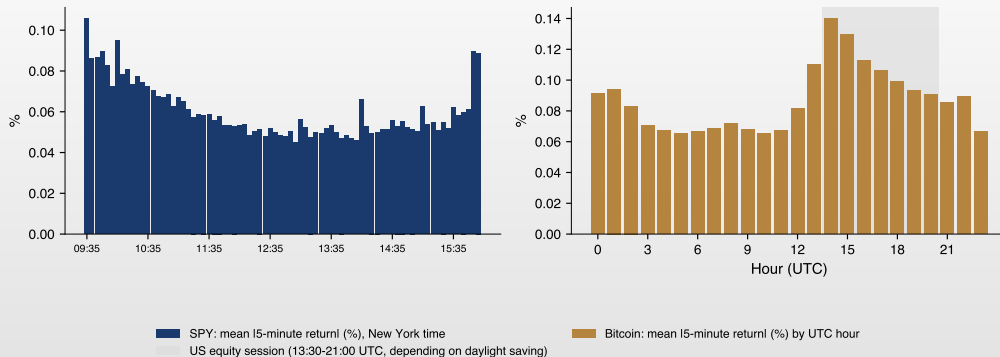


□ SPY, March–May 2025; band: 95% confidence interval for  $\sqrt{252 IV_t}$  from realised quarticity

## Reading the Confidence Intervals

- Median relative width of the interval (for volatility): 37% of its level
  - ▶ The band assumes a continuous price and negligible noise: on a jump day RV tends to  $IV_t + \sum_j \kappa_j^2$ , so it covers total variation, not  $IV_t$  (Section 4)
- 9 April 2025 (the day the tariff pause was announced): volatility 72%, CI [55%, 93%]
  - ▶ The interval widens when a few returns dominate the sum:  $\sum r^4$  grows faster than  $(\sum r^2)^2$
- Still far more precise than  $r_t^2$ , whose relative error is of order  $\sqrt{2} \approx 141\%$  under the Normal distribution
- Practical rule: a single day's volatility is known to about  $\pm 20\%$  (95% band), averages over weeks much more precisely

## Intraday Seasonality



□ Left: SPY, mean  $|r_{t,i}|$  by 5-minute interval of the day (full days only); right: Bitcoin by UTC hour

## Reading the Seasonality Pattern

- SPY: a **U-shape**: 0.106% in the first 5 minutes, 0.050% around midday, 0.089% in the last 5 minutes [Andersen & Bollerslev \(1997\)](#)
  - ▶ The open prices overnight news; the close concentrates index and fund rebalancing
- Bitcoin trades 24/7 but is most volatile at 14:00 UTC, near the US equity opening (09:30 New York: 13:30 UTC in summer, 14:30 UTC in winter); quietest at 10:00 UTC (ratio 2.2)
- RV sums over the day, so seasonality does not bias it; it does bias BV:  $\mathbb{E}|r_i| |r_{i-1}| \propto \sigma_i \sigma_{i-1} \leq (\sigma_i^2 + \sigma_{i-1}^2)/2$ , so fast changes of scale push RV – BV up and the jump test (Section 4) over-rejects
  - ▶ Remedy: divide  $r_{t,i}$  by a periodicity factor  $f_i$  (standard-deviation estimator of [Boudt, Croux & Laurent \(2011\)](#)):  $f = 1.78$  at the open, 0.73 at the minimum; SPY rejections at 5%: 235  $\rightarrow$  167

## Recap: Realised Variance

- ▣  $RV_t = \sum_i r_{t,i}^2$  measures integrated variance without a model; precision improves with  $M$
- ▣ Realised quarticity gives its standard error: about  $\sqrt{2/78} = 16\%$  of RV for one day of 5-minute data, i.e. a 95% band of roughly  $\pm 20\%$  for volatility
- ▣ For SPY, overnight returns add 38% of daily variance: use total variance for daily risk
- ▣ Volatility follows a daily U-shape (SPY) and a UTC-hour pattern (Bitcoin)

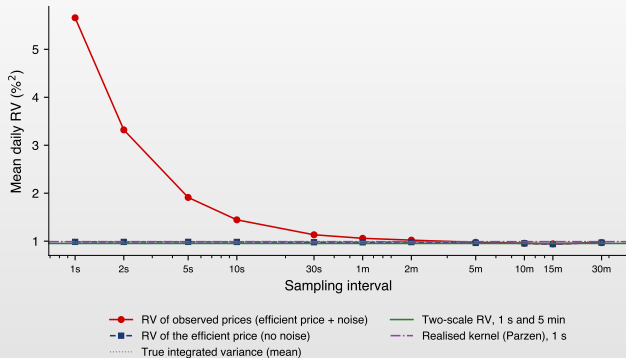
## Why Not Use Every Tick?

- Observed prices are not the efficient price:  $p_i^{\text{obs}} = p_i^* + \varepsilon_i$ 
  - ▶ Sources of  $\varepsilon$ : bid–ask bounce, tick size (price discreteness), stale quotes (Chapter 10)
- With i.i.d. noise of variance  $\omega^2$  and  $n$  returns per day [Zhang, Mykland & Aït-Sahalia \(2005\)](#), [Hansen & Lunde \(2006\)](#)
  - ▶  $\mathbb{E}[\text{RV}_t^{(n)} \mid \sigma] = \text{IV}_t + 2n\omega^2$ : the bias **grows** with the sampling frequency
  - ▶ Observed returns become negatively autocorrelated:  $\text{cov}(r_i, r_{i-1}) = -\omega^2$
- Trade-off: more returns reduce sampling error but add noise bias

## Deriving the Noise Bias and the Optimal Frequency

- Observed return  $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-1}$ , with  $\varepsilon$  i.i.d.  $(0, \omega^2)$  and independent of  $p^*$ 
  - ▶ Given the path of  $\sigma$  (zero drift):  $\mathbb{E}[\text{RV}^{(n)} \mid \sigma] = \sum_i \mathbb{E}[r_i^{*2} \mid \sigma] + \sum_i \mathbb{E}[(\varepsilon_i - \varepsilon_{i-1})^2] = \text{IV} + 2n\omega^2$ ;  
 $\text{cov}(r_i, r_{i-1}) = -\mathbb{E}[\varepsilon_{i-1}^2] = -\omega^2$
  - ▶ By a law of large numbers for the noise terms,  $\text{RV}^{(n)}/(2n) \xrightarrow{p} \omega^2$  as  $n \rightarrow \infty$ : the all-tick RV estimates the noise, not IV  
[Bandi & Russell \(2008\)](#)
- MSE:  $\mathbb{E}[(\text{RV}^{(n)} - \text{IV})^2] \approx \underbrace{2\text{IQ}/n}_{\text{discretisation}} + \underbrace{4n^2\omega^4}_{\text{squared bias}} + O(n)$ 
  - ▶ First-order condition  $-2\text{IQ}/n^2 + 8n\omega^4 = 0$  gives  $n^* = (\text{IQ}/(4\omega^4))^{1/3}$  [Bandi & Russell \(2008\)](#)
- Numbers:  $\text{IV} = 1\%^2$ ,  $\text{IQ} = 1$ ,  $\omega = 0.01\%$ : bias 0.016 at 5 minutes, 4.680 at 1 second;  $n^* = 292$  (one return every 80 s)

## The Volatility Signature Plot: Simulation



100 simulated days, one price per second; efficient price with random daily volatility plus i.i.d. noise,  $\omega = 0.01\%$

## Reading the Simulated Signature Plot

- True mean IV = 0.98; RV of observed prices: 5.66 at 1 second, 1.06 at 1 minute, 0.98 at 5 minutes
  - ▶ The bias matches  $2n\omega^2$  from the previous slide
- Without noise, RV is flat in the sampling interval: the rise at high frequency is the fingerprint of noise
- Noise-robust estimators on 1-second data: two-scale RV 0.95, realised kernel 0.99
  - ▶ Error standard deviation across days: 5-minute RV 0.187, two-scale 0.150, kernel 0.045: using all data pays off only with a noise correction

## Remedies: Sparse Sampling and Subsampling

- **Sparse sampling:** use one return every 5 minutes instead of every tick
  - ▶ Simple and robust; throws away most of the data [Liu, Patton & Sheppard \(2015\)](#)
- **Subsampling:** compute RV on  $K$  shifted grids (offsets  $0, 1, \dots, K - 1$ ) and average
  - ▶  $RV_t^{\text{avg}} = \frac{1}{K} \sum_{k=0}^{K-1} RV_t^{(k)}$ : same bias as one sparse grid, smaller variance
- **Two-scale RV (TSRV)** [Zhang, Mykland & Aït-Sahalia \(2005\)](#): remove the remaining bias with the all-data RV
  - ▶  $TSRV_t = RV_t^{\text{avg}} - \frac{\bar{n}}{n} RV_t^{\text{all}}$ ,  $\bar{n} = (n - K + 1)/K$ ; divide by  $1 - \bar{n}/n$  in small samples
  - ▶  $\bar{n}$  is the average number of returns per sparse grid; our grids also keep the open and the close, so  $\bar{n}$  is their actual average count

## Realised Kernels



Neil Shephard, 2004

Photo: Renate Schmid (2004); CC BY-SA  
2.0 de; Wikimedia Commons

- ▣ **Realised kernel** Barndorff-Nielsen, Hansen, Lunde & Shephard (2008): add weighted autocovariances of returns
  - ▶  $RK_t = \gamma_0 + 2 \sum_{h=1}^H k\left(\frac{h}{H+1}\right) \gamma_h$ ,  $\gamma_h = \sum_i r_{t,i} r_{t,i-h}$
  - ▶ Parzen weight function  $k$ : 1 at 0, falling smoothly to 0 at 1; keeps RK positive
- ▣ Noise makes  $\gamma_1$  negative; the kernel adds it back and cancels the bias
- ▣ Bandwidth  $H$  grows with the noise-to-signal ratio:  $H \approx 3.5 \xi^{4/5} n^{3/5}$ ,  $\xi^2 = \omega^2 / \sqrt{IQ}$
- ▣ Shephard and co-authors developed kernels, bipower variation and the asymptotic theory of RV

## Rates of Convergence under Noise

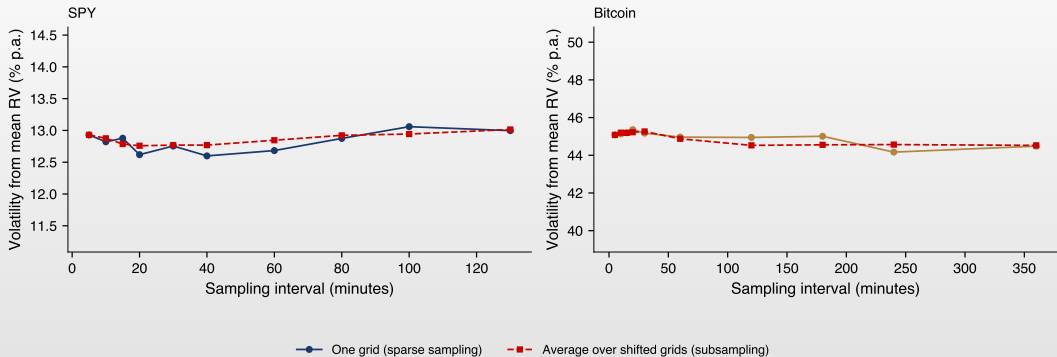
| Estimator                          | Error of order                     | Positive | Source                                             |
|------------------------------------|------------------------------------|----------|----------------------------------------------------|
| RV on all ticks                    | inconsistent (bias $2n\omega^2$ )  | yes      | Bandi & Russell (2008)                             |
| Sparse RV, fixed 5-minute grid     | $M^{-1/2}$ , $M$ fixed; bias stays | yes      | Liu, Patton & Sheppard (2015)                      |
| Two-scale RV                       | $n^{-1/6}$                         | no       | Zhang, Mykland & Aït-Sahalia (2005)                |
| Multi-scale RV                     | $n^{-1/4}$                         | no       | Zhang (2006)                                       |
| Realised kernel, flat-top / Parzen | $n^{-1/4}$ / $n^{-1/5}$            | no / yes | Barndorff-Nielsen, Hansen, Lunde & Shephard (2008) |
| Pre-averaging                      | $n^{-1/4}$                         | no       | Jacod, Li, Mykland, Podolskij & Vetter (2009)      |

- $n^{-1/4}$  is the optimal rate with i.i.d. noise: the price of noise is half the usual exponent
- With  $n = 23\,400$  seconds:  $n^{1/4} \approx 12.4$ , against  $\sqrt{78} \approx 8.8$  for 5-minute RV: at best a modest gain, which explains Liu, Patton & Sheppard (2015)

## Pre-Averaging

- Local averages of  $k_n \approx \theta\sqrt{n}$  returns with a weight  $g$  on  $[0, 1]$ , e.g.  $g(x) = \min(x, 1 - x)$  Jacod, Li, Mykland, Podolskij & Vetter (2009)
  - ▶  $\bar{r}_i = \sum_{j=1}^{k_n-1} g(j/k_n) r_{i+j}$ : noise variance shrinks like  $1/k_n$ , the signal keeps order  $k_n/n$
- Estimator, with  $\psi_1 = \int_0^1 g'(x)^2 dx$ ,  $\psi_2 = \int_0^1 g(x)^2 dx$  (1 and 1/12 for the triangular  $g$ )
  - ▶  $\widehat{IV} = \frac{1}{\theta \psi_2 \sqrt{n}} \sum_i \bar{r}_i^2 - \frac{\psi_1}{2 \theta^2 \psi_2 n} \sum_i r_i^2$  (second term: noise bias correction)
- Rate  $n^{-1/4}$ ; the same device gives jump-robust (pre-averaged bipower) and covariance estimators
- Built for tick data: on 5-minute bars little noise is left to average out (next slides)

## The Signature Plot on Real Data



- Annualised volatility from mean RV by sampling interval: one grid vs the average over all shifted grids

## Reading the Empirical Signature Plot

- SPY: 12.9% at 5 minutes, 12.8% at 30 minutes, 13.0% at 130 minutes: flat
  - ▶ First-order autocorrelation of 5-minute returns:  $-0.008$  (SPY),  $0.005$  (Bitcoin): no visible noise at 5 minutes
- Bitcoin: 45.1% at 5 minutes, 44.5% at 6 hours: flat as well
- On 5-minute SPY data: two-scale  $RV / RV = 0.97$ , realised kernel /  $RV = 0.95$ ; correlations with RV 0.89 and 0.85
  - ▶ The diagnostics show little noise bias at 5 minutes: the corrections move RV by a few per cent and add estimation variability
- Consistent with [Liu, Patton & Sheppard \(2015\)](#): almost 400 estimators on 31 assets rarely beat 5-minute RV; noise-robust estimators pay off when noise is large relative to the signal (tick or second-level data, illiquid assets)

## Realised Covariance and Asynchronous Trading

- Realised covariance  $\sum_i r_{t,i}^{(1)} r_{t,i}^{(2)} \rightarrow \int \sigma_{12}(s) ds$  needs prices of both assets on the **same** grid
  - ▶ Epps effect [Epps \(1979\)](#): measured correlation falls towards 0 as sampling gets finer, because trades are asynchronous and the last price is stale
- [Hayashi & Yoshida \(2005\)](#): sum the products of **all** pairs of returns whose time intervals overlap
  - ▶  $\sum_{i,j} r_i^{(1)} r_j^{(2)} 1\{(t_{i-1}, t_i] \cap (s_{j-1}, s_j] \neq \emptyset\}$ : unbiased without synchronisation (in the absence of noise)
- With noise as well: pre-averaged or multivariate kernel versions; application: intraday portfolio risk (Chapter 4)

## Recap: Microstructure Noise

- ▣ Noise adds  $2n\omega^2$  to RV: sampling too often overstates volatility
- ▣ The signature plot is the diagnostic: RV should not depend on the sampling interval
- ▣ Remedies: 5-minute sampling, subsampling, two-scale RV ( $n^{-1/6}$ ), realised kernels and pre-averaging ( $n^{-1/4}$ )
- ▣ For liquid assets at 5 minutes (SPY, Bitcoin) plain RV is the benchmark

## Prices Jump

- Add jumps to the model:  $dp = \mu ds + \sigma dW + dJ$ , with  $J$  a jump process (news, Chapter 11)
  - ▶ Quadratic variation now has two parts:  $RV_t \rightarrow IV_t + \sum_j \kappa_j^2$ , with  $\kappa_j$  the jump sizes in day  $t$
- Why separate them? Jumps are less persistent than diffusive volatility, and they price differently (options, Chapter 12)
- We need an estimator of  $IV_t$  that ignores jumps

## Bipower Variation



Ole E. Barndorff-Nielsen, 2007

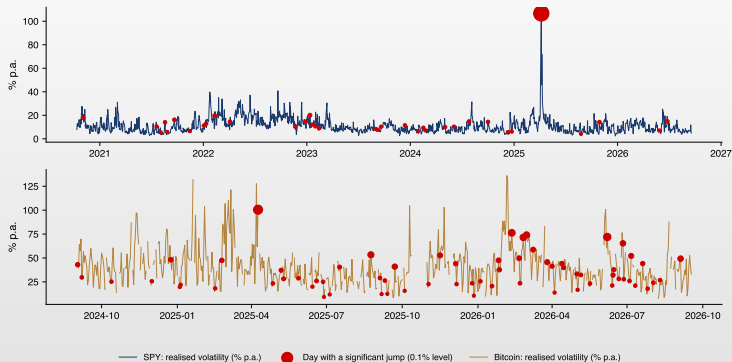
Photo: Thomas Steiner (2007); CC BY-SA 2.5;  
Wikimedia Commons

- **Bipower variation** Barndorff-Nielsen & Shephard (2004): products of adjacent absolute returns
  - ▶  $BV_t = \mu_1^{-2} \frac{M}{M-1} \sum_{i=2}^M |r_{t,i}| |r_{t,i-1}| \rightarrow IV_t$
  - ▶  $\mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi} = 0.798$  for  $Z \sim N(0, 1)$ , so  $\mu_1^{-2} = \pi/2 = 1.571$
- A jump enters one return: it is multiplied by a small neighbour, so its effect vanishes as  $M$  grows
- Jump part of the day:  $\max(RV_t - BV_t, 0)$
- Barndorff-Nielsen (1935–2022) also shaped the Lévy-process models used for jumps

## A Test for Jumps

- ▣ Ratio statistic [Huang & Tauchen \(2005\)](#), [Barndorff-Nielsen & Shephard \(2006\)](#): 
$$z_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{\theta M^{-1} \max(1, TQ_t/BV_t^2)}}$$
  - ▶  $\theta = \pi^2/4 + \pi - 5 = 0.609$ ; tripower quarticity  $TQ_t = M\mu_{4/3}^{-3} \frac{M}{M-2} \sum_i |r_{t,i}|^{4/3} |r_{t,i-1}|^{4/3} |r_{t,i-2}|^{4/3}$ ,  $\mu_{4/3} = \mathbb{E}|Z|^{4/3}$
- ▣ Under no jumps  $z_t \rightarrow N(0, 1)$ ; a jump makes  $RV > BV$  and  $z_t$  large and positive
- ▣ One test per day: at level  $\alpha$  we expect  $\alpha \times$  days false alarms
  - ▶ We use  $\alpha = 0.1\%$  (critical value 3.09): about 1.5 false jump days for SPY
- ▣ Significant day: jump part  $J_t = RV_t - BV_t$ , continuous part  $C_t = RV_t - J_t$ ; otherwise  $J_t = 0$

## Jump Days: SPY and Bitcoin



Red points: days with a significant jump at the 0.1% level; point size proportional to the jump part  $J_t$

## Reading the Jump Test

- SPY: 35 jump days of 1 489 (2.4%); jumps are 2.8% of total realised variance
  - ▶ On a jump day the jump is on average 35% of that day's RV; BV/RV averages 0.95 over all days
  - ▶ Largest statistic: 2021-08-27 ( $z = 5.2$ ), jump in the 10:00–10:05 interval, when the Federal Reserve chair's Jackson Hole speech was released
- Bitcoin: 67 jump days of 659 (10.2%), but only 1.8% of variance
  - ▶ Many small jumps on quiet days (weekends): one isolated move dominates a calm day
- The turbulent days of April 2025 are mostly **not** jump days: many large returns in a row raise BV as well

## Many Days, Many Tests: Spurious Jumps

- 1 489 daily tests at 5%: 235 rejections against 74 expected under no jumps; which of them are false?
- [Benjamini & Hochberg \(1995\)](#): sort the  $p$ -values, reject the  $k$  smallest, with  $k$  the largest index such that  $p_{(k)} \leq kq/N$ ; the FDR,  $\mathbb{E}[\text{false/rejections}]$ , stays  $\leq q$  (with false/rejections  $:= 0$  when nothing is rejected)
  - ▶ The guarantee needs valid  $p$ -values that are independent or positively dependent [Benjamini & Yekutieli \(2001\)](#); daily jump tests are only approximately so
  - ▶ SPY: FDR at 5%: 42 jump days; at 1%: 9; Bonferroni (FWER 5%): 7; FDR 5% after the periodicity correction: 17
- [Bajgrowicz, Scaillet & Treccani \(2016\)](#): without multiplicity control most detected jumps are spurious; the true ones are rare and cluster on news
- [Christensen, Oomen & Podolskij \(2014\)](#): at tick frequency most 5-minute “jumps” are steep but continuous bursts of volatility (fact or friction)

## Locating Jumps Within the Day: Lee–Mykland

- Lee & Mykland (2008):  $L_{t,i} = r_{t,i} / \hat{\sigma}_{t,i}$ , with  $\hat{\sigma}_{t,i}^2$  the bipower average of the previous  $K = 270$  returns (their value for 5-minute data)
  - ▶ Under no jumps,  $\max_i |L_i|$  over  $n$  returns has a Gumbel limit: reject if  $(|L| - C_n) / S_n > -\ln(-\ln(1 - \alpha))$ , with  $C_n = \sqrt{2 \ln n} - \frac{\ln \pi + \ln \ln n}{2\sqrt{2 \ln n}}$  and  $S_n = 1 / \sqrt{2 \ln n}$  (our  $\hat{\sigma}^2$  includes the factor  $\pi/2$ , so the constant  $c = \sqrt{2/\pi}$  of Lee & Mykland (2008) becomes 1)
  - ▶ SPY, returns divided by  $f_i$ ,  $\alpha = 1\%$ ,  $n = 115\,453$  (the whole sample): threshold  $|L| > 5.41$
- 169 jumps on 105 days; 62 of them on Wednesdays between 14:00 and 15:00, when FOMC statements (14:00) and press conferences (14:30) are released
  - ▶ Largest:  $L = -22.9$  on 2025-10-29; only 10% of these days are also flagged by the daily ratio test
- The two tests ask different questions: one extreme return relative to local volatility, versus a daily gap between RV and BV

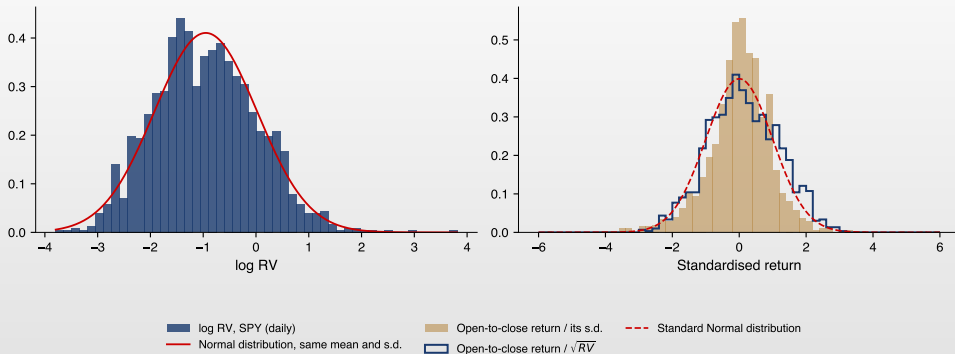
## Jump-Robust Estimators of Integrated Variance

- Truncated RV [Mancini \(2009\)](#):  $\text{TRV}_t = \sum_i r_{t,i}^2 \mathbf{1}\{|r_{t,i}| \leq c \hat{\sigma}_t f_i \Delta^\varpi\}$ ,  $\varpi \in (0, 1/2)$ 
  - ▶ Diffusive increments are  $O_p(\Delta^{1/2})$  and end up below the threshold; jumps are  $O(1)$  and are cut off
- MedRV [Andersen, Dobrev & Schaumburg \(2012\)](#):  $\frac{\pi}{6-4\sqrt{3}+\pi} \frac{M}{M-2} \sum_i \text{med}(|r_{i-1}|, |r_i|, |r_{i+1}|)^2$ 
  - ▶ The median drops an isolated jump entirely, even in finite samples, and is less sensitive than BV to isolated zero returns; two adjacent jumps still leak in (the median is then the smaller jump)
- Jump share of SPY RV: BV 6.3%, MedRV 7.2%, TRV 12.5% ( $c = 3$ ,  $\varpi = 0.49$ ; truncation bites on 44% of days:  $c$  is a tuning choice)
- [Aït-Sahalia & Jacod \(2009\)](#): compare power variations  $\sum |r|^p$ ,  $p > 2$ , at  $\Delta$  and  $k\Delta$ ; the ratio tends to 1 with jumps and to  $k^{p/2-1}$  without: known limits under both hypotheses

## Recap: Jumps

- RV measures total quadratic variation: diffusion plus squared jumps
- Bipower variation is robust to jumps;  $RV - BV$  estimates the jump part
- The ratio test flags about 2.4% of SPY days at the 0.1% level; jumps are a small share of total variance
- Test many days, control the FDR: 235 SPY days at 5% shrink to 42 with Benjamini–Hochberg; Lee–Mykland locates jumps in time (FOMC releases)
  - ▶ Jumps are separated from continuous variation in the HAR-CJ model (Section 5, Seminar 9)

## Distributions: log RV and Standardised Returns

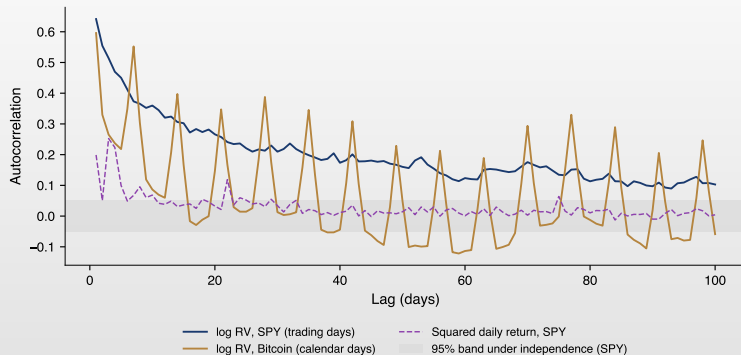


- Left: daily log RV of SPY and the Normal density; right: open-to-close returns scaled by their standard deviation (filled) and by  $\sqrt{RV_t}$  (outline)

## Reading the Distributions

- SPY RV: skewness 20.0, kurtosis 550; log RV: skewness 0.29, kurtosis 3.20 (the Normal distribution: 0 and 3)
  - ▶ Bitcoin log RV: skewness  $-0.27$ , kurtosis 3.33: model volatility in logs
- Open-to-close returns: kurtosis 15.73; divided by  $\sqrt{RV_t}$ : 2.65, standard deviation 1.05
  - ▶ Bitcoin: from 7.57 to 2.77
- Heavy tails of daily returns (Chapter 1) come mostly, ex post, from changing volatility: returns are a **mixture** of Normal distributions with variances  $IV_t$

## Long Memory in Realised Volatility

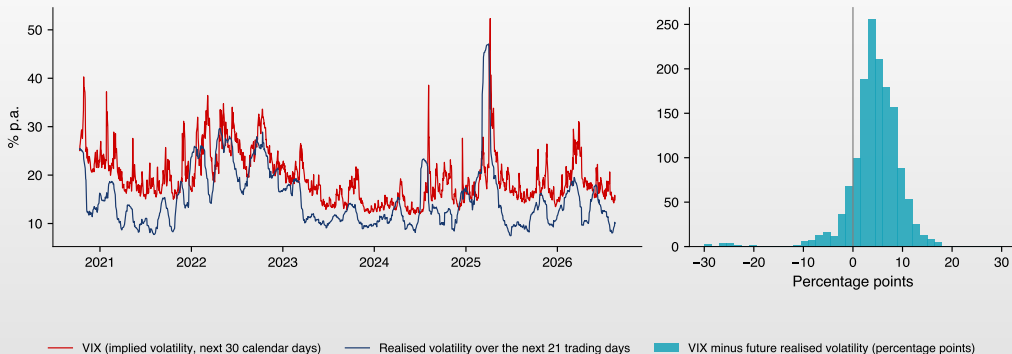


- Autocorrelations up to 100 days: log RV of SPY (total variance) and Bitcoin, squared daily SPY returns

## Reading the Autocorrelations

- SPY log RV: 0.64 at lag 1, 0.24 at one month, still 0.10 at 100 days: **long memory**
  - ▶ An AR(1) would decay geometrically:  $0.64^{22} \approx 0.00005$
- Squared daily returns show the same persistence, hidden by noise: 0.20 at lag 1
- Bitcoin: peaks at 7, 14, 21, ... days: a **weekly cycle**
  - ▶ Weekend RV is only 0.41 of weekday RV (Saturday 27% vs Wednesday 48% p.a.): no equity or macro news, fewer institutional traders

## Implied vs Realised Volatility



- Left: VIX and SPY realised volatility over the next 21 trading days (including overnight returns); right: the gap between them

## Reading the Variance Risk Premium

- VIX: the market's 30-day volatility, implied by S&P 500 options (Chapter 12); mean 19.5% vs 15.2% realised afterwards
  - ▶ VIX exceeded the subsequent realised volatility on 88% of 1 468 days; mean gap 4.4 percentage points
- **Variance risk premium (VRP)** Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009):  
$$\text{VRP}_t = \mathbb{E}_t^Q[\text{RV}_{t,t+21}] - \mathbb{E}_t^P[\text{RV}_{t,t+21}]$$
, risk-neutral minus physical expectation of the same future variance;  $\text{VIX}^2$  approximates the first term
  - ▶ Investors pay to insure against volatility spikes; the chart shows an **ex-post** spread (VIX minus the volatility that followed), a noisy proxy of this ex-ante premium
- As a forecast: regressing future variance on  $\text{VIX}^2$  gives slope 0.56 (SE 0.05),  $R^2 = 0.23$ : VIX is informative but biased upwards
- The ex-post spread turns negative when realised volatility exceeds the earlier implied volatility (March–April 2025): a forecast error, not evidence of a negative ex-ante premium

## Recap: Stylised Facts

- log RV is close to Normal Andersen, Bollerslev, Diebold & Labys (2001), Andersen, Bollerslev, Diebold & Ebens (2001): model and forecast volatility in logs
- Returns divided by same-day  $\sqrt{RV}$  are close to  $N(0, 1)$  ex post: fat tails come mostly from changing volatility
- Long memory: the model must weigh the last month, not only yesterday
- Implied volatility carries a risk premium: it is an upward-biased forecast

## The HAR-RV Model

- Corsi (2009):  $RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}$ 
  - ▶  $RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}$  (week),  $RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$  (month)
  - ▶ Bitcoin trades every day: week = 7 days, month = 30 days
- Heterogeneous market hypothesis Müller et al. (1997): daily, weekly and monthly traders; a few components with different persistence mimic long memory
- An AR(22) with only 3 free slopes: weight  $\beta_d + \beta_w/5 + \beta_m/22$  on lag 1,  $\beta_w/5 + \beta_m/22$  on lags 2–5,  $\beta_m/22$  on lags 6–22
- Estimation by OLS; errors are heteroskedastic and autocorrelated: use Newey–West (HAC) standard errors
  - ▶ log-HAR: the same regression for  $\ln RV$ ; forecast  $\exp(\hat{y} + \hat{s}^2/2)$ ,  $\hat{s}^2$  the residual variance: the conditional mean if the errors are  $N(0, s^2)$  (lognormal approximation)
  - ▶ For non-Normal errors use the smearing factor  $\frac{1}{T} \sum_t e^{\hat{\varepsilon}_t}$  Duan (1983); with heteroskedastic errors both are approximations

## Extensions of HAR

- ▣ **HAR-CJ Andersen, Bollerslev & Diebold (2007)**: separate continuous and jump parts as regressors
  - ▶  $\ln RV_{t+1} = \beta_0 + \beta_{Cd} \ln C_t + \beta_{Cw} \ln C_t^{(w)} + \beta_{Cm} \ln C_t^{(m)}$
  - ▶  $+ \beta_{Jd} \ln(1 + J_t) + \beta_{Jw} \ln(1 + J_t^{(w)}) + \beta_{Jm} \ln(1 + J_t^{(m)}) + \varepsilon_{t+1}$
  - ▶  $C_t^{(w)}, J_t^{(w)}$  ( $C_t^{(m)}, J_t^{(m)}$ ): arithmetic means over the last 5 (22) days, as  $RV_t^{(w)}$ ; logs are taken after averaging
- ▣ **HARQ Bollerslev, Patton & Quaedvlieg (2016)**: less weight on yesterday's RV when it is imprecise (high realised quarticity)
  - ▶  $\beta_d$  becomes  $\beta_d + \beta_Q \sqrt{RQ_t}$  with  $\beta_Q < 0$
- ▣ Weekend or holiday dummies, leverage (negative returns raise future RV), implied volatility as an extra regressor
- ▣ Multi-step: regress  $\sum_{j=1}^h RV_{t+j}$  directly on the same regressors

## HAR Estimates: SPY and Bitcoin

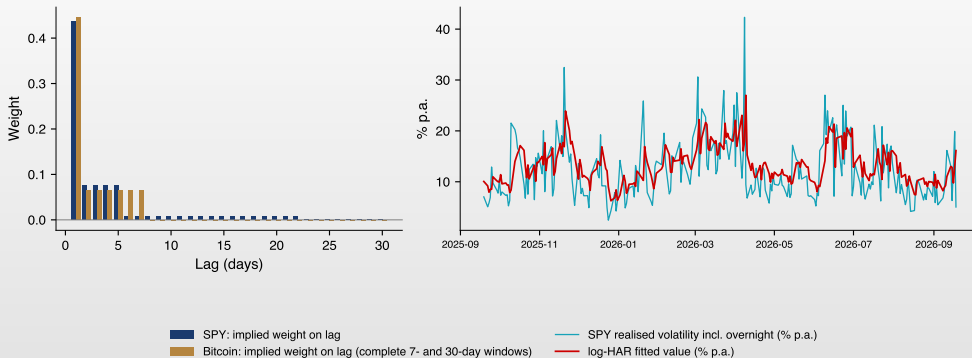
|                       | $\beta_0$         | $\beta_d$        | $\beta_w$        | $\beta_m$         | $R^2$ |
|-----------------------|-------------------|------------------|------------------|-------------------|-------|
| SPY, HAR (levels)     | 0.273<br>(0.064)  | 0.399<br>(0.089) | 0.222<br>(0.096) | 0.114<br>(0.053)  | 0.31  |
| SPY, log-HAR          | -0.079<br>(0.027) | 0.362<br>(0.037) | 0.337<br>(0.048) | 0.167<br>(0.044)  | 0.47  |
| Bitcoin, HAR (levels) | 1.496<br>(0.412)  | 0.412<br>(0.089) | 0.409<br>(0.122) | -0.065<br>(0.097) | 0.34  |
| Bitcoin, log-HAR      | 0.223<br>(0.094)  | 0.380<br>(0.041) | 0.465<br>(0.080) | -0.022<br>(0.098) | 0.36  |

- Full samples; SPY: total daily variance (intraday RV plus squared overnight return), 1 467 days; Bitcoin: 24-hour RV, 547 days
- Newey–West standard errors in brackets

## Reading the HAR Estimates

- SPY: all three components are significant; in logs  $\hat{\beta}_d = 0.362$ ,  $\hat{\beta}_w = 0.337$ ,  $\hat{\beta}_m = 0.167$ 
  - ▶ Persistence  $\hat{\beta}_d + \hat{\beta}_w + \hat{\beta}_m = 0.87$ : shocks fade slowly, as in GARCH (Chapter 5)
- $R^2 = 0.47$  for log variance vs 0.31 for variance in levels: not comparable, the dependent variables differ; the out-of-sample losses on a common target (Section 7) decide
- Bitcoin: daily and weekly components matter; the monthly coefficient is close to 0 ( $-0.022$ , SE 0.098)
  - ▶ Short sample (two years) and a strong weekly cycle: the week absorbs most of the memory

## Implied Lag Weights and In-Sample Fit



- Left: weights of log-HAR on each past day (Bitcoin: complete 7- and 30-day windows); right: SPY, last 250 days, realised volatility and log-HAR fitted values

## Reading the Lag Weights

- SPY: yesterday gets 0.437, each of days 2–5 gets 0.075, each of days 6–22 gets 0.008
- The staircase decays slowly: the month still holds weight  $22 \times 0.008 \approx 0.167$ 
  - ▶ Three coefficients reproduce the long memory of the autocorrelation function
- Bitcoin: yesterday 0.446, days 2–7 about 0.066 each, almost nothing beyond a week
- In-sample fitted values follow the level of volatility, not the day-to-day noise

## HAR as a Restricted AR(22): Testing the Restrictions

- Unrestricted:  $\ln RV_{t+1} = \phi_0 + \sum_{k=1}^{22} \phi_k \ln RV_{t+1-k} + \varepsilon_{t+1}$ 
  - ▶ HAR imposes  $\phi_2 = \dots = \phi_5$  (3 restrictions) and  $\phi_6 = \dots = \phi_{22}$  (16):  $R\phi = 0$  with 19 rows
- Wald:  $W = (R\hat{\phi})^\top (R\hat{V}_{\text{NW}}R^\top)^{-1} R\hat{\phi} \rightarrow \chi^2_{19}$  under the HAR restrictions
- SPY log total variance:  $W = 18.1$ ,  $p = 0.51$ ;  $R^2$  0.477 (AR(22)) vs 0.470 (HAR)
  - ▶ The staircase is not rejected: HAR saves 19 parameters at almost no in-sample cost, and less estimation noise helps out of sample

## Measurement Error in the Regressors: Why HARQ

- $RV_t = IV_t + u_t$  with  $\mathbb{E}[u_t | \sigma] = 0$  and  $\text{Var}(u_t | \sigma) = 2IQ_t/M$  (the CLT of Section 2)
  - ▶ If  $IV_{t+1} = \beta_0 + \beta IV_t + \eta_{t+1}$ , OLS of  $RV_{t+1}$  on  $RV_t$  gives  $\text{plim } \hat{\beta} = \beta \frac{\text{Var}(IV)}{\text{Var}(IV) + \mathbb{E}[2IQ/M]} < \beta$  for  $\beta > 0$ : attenuation
- If the signal variance does not rise with  $IQ_t$  (a working approximation), the signal-to-noise ratio of  $RV_t$  falls when  $IQ_t$  is high; HARQ approximates the falling weight by  $\beta_d + \beta_Q \sqrt{RQ_t}$ ,  $\beta_Q < 0$  [Bollerslev, Patton & Quaedvlieg \(2016\)](#)
- SPY, levels, Newey–West SE: intraday RV  $\hat{\beta}_Q = -0.0088$  (0.0042); total variance  $-0.0022$  (0.0024)
  - ▶ Daily slope (intraday RV) 1.13 at the median  $\sqrt{RQ}$ , 1.09 at its 99th percentile: the sign predicted by the theory, a small effect

## Recap: HAR-RV

- HAR regresses tomorrow's RV on daily, weekly and monthly averages: OLS, three slopes
- It approximates long memory with a staircase of lag weights
- Estimate in logs; report Newey–West standard errors
- HAR is an AR(22) with 19 restrictions that the SPY data do not reject; measurement error in  $RV_t$  attenuates the daily slope, which HARQ corrects approximately
- Extensions: jumps (HAR-CJ), measurement error (HARQ), calendar effects, implied volatility

## Forecasting Design

- Target: tomorrow's daily variance; SPY: total variance, 2022-01-03 to 18 September 2026 (1 180 days); Bitcoin: 2025-03-01 onwards (422 days)
  - ▶ Every forecast uses only data available the evening before
- Five models
  - ▶ log-HAR and HAR on realised variance, re-estimated daily on an expanding window
  - ▶ GARCH(1,1) with Student- $t$  errors on daily returns since 2000 (SPY) or 2014 (Bitcoin), re-estimated every 21 days [Bollerslev \(1986\)](#)
  - ▶ EWMA with  $\lambda = 0.94$  (Chapter 5) and "yesterday's RV" (random walk)
- HAR sees intraday data; GARCH and EWMA see only daily returns

## Evaluating Against a Noisy Target

- True variance is unobserved; we compare forecasts  $F_t$  with the proxy  $RV_t$ 
  - ▶ A loss is **robust** if the ranking with a noisy unbiased proxy equals the ranking with true variance [Patton \(2011\)](#)
- Two robust losses
  - ▶  $MSE = (RV_t - F_t)^2$ : dominated by the most volatile days
  - ▶  $QLIKE = \frac{RV_t}{F_t} - \ln \frac{RV_t}{F_t} - 1 \geq 0$ : depends only on the ratio; penalises under-prediction more
- Common mistake: MAE, or MSE on volatility instead of variance, is **not** robust
  - ▶ [Andersen & Bollerslev \(1998\)](#): against  $r_t^2$ , GARCH forecasts explained a few % of the variation; against RV the same forecasts explain far more: the proxy was noisy, not the model

## Tests: Diebold–Mariano and Mincer–Zarnowitz

- Diebold & Mariano (1995): loss difference  $d_t = L(RV_t, F_t^A) - L(RV_t, F_t^B)$ ;  $t = \bar{d}/\text{se}_{\text{NW}}(\bar{d}) \rightarrow N(0, 1)$ 
  - ▶  $t < 0$ : model A has the lower loss; HAC standard error because  $d_t$  is autocorrelated (Chapter 8)
- Mincer & Zarnowitz (1969):  $RV_t = a + b F_t + u_t$ ; an unbiased forecast has  $a = 0$ ,  $b = 1$ 
  - ▶ Joint Wald test with Newey–West covariance;  $R^2$  measures how much of the variation the forecast explains
  - ▶  $R^2$  is capped by the noise in RV: even the true variance would not reach 1 Andersen & Bollerslev (1998)

## Results: SPY, 2022–2026

| Model           | QLIKE | DM vs log-HAR ( $p$ ) | MSE  | MZ $b$ | MZ $R^2$ |
|-----------------|-------|-----------------------|------|--------|----------|
| log-HAR         | 0.310 | –                     | 4.15 | 1.16   | 0.30     |
| HAR             | 0.326 | –1.95 (0.052)         | 4.48 | 0.75   | 0.26     |
| GARCH(1,1)- $t$ | 0.343 | –2.30 (0.021)         | 4.87 | 0.70   | 0.20     |
| EWMA            | 0.393 | –4.02 ( $< 0.001$ )   | 5.14 | 0.74   | 0.14     |
| Yesterday's RV  | 0.502 | –5.95 ( $< 0.001$ )   | 5.41 | 0.54   | 0.29     |

□ 1180 one-day-ahead forecasts of total daily variance; DM: QLIKE differences, Newey–West;  $t < 0$  favours log-HAR

## Reading the SPY Results

- log-HAR has the lowest QLIKE and MSE: QLIKE 10% below GARCH(1,1)- $t$  (DM  $t = -2.30$ ,  $p = 0.021$ )
  - ▶ Against EWMA and yesterday's RV the gain is highly significant; against HAR in levels only marginally ( $p = 0.052$ )
- MSE differences are not significant: a few crisis days (April 2025) dominate the squared errors
- Mincer–Zarnowitz: log-HAR  $b = 1.16$  ( $p = 0.554$  for  $a = 0$ ,  $b = 1$ ), GARCH  $b = 0.70$ : no significant bias, but log-HAR explains more ( $R^2$  0.30 vs 0.20)
- Yesterday's RV has a high  $R^2$  but  $b = 0.54$ : right direction, far too extreme

## Results: Bitcoin, 2025–2026

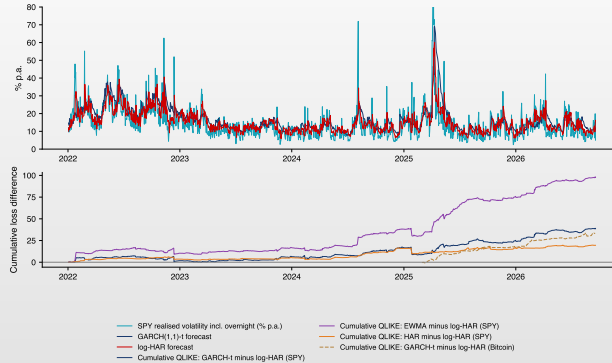
| Model           | QLIKE | DM vs log-HAR ( $p$ ) | MSE  | MZ $b$ | MZ $R^2$ |
|-----------------|-------|-----------------------|------|--------|----------|
| log-HAR         | 0.282 | –                     | 26.6 | 0.82   | 0.38     |
| HAR             | 0.409 | –1.22 (0.223)         | 27.6 | 0.89   | 0.34     |
| GARCH(1,1)- $t$ | 0.360 | –4.32 ( $< 0.001$ )   | 42.2 | 0.55   | 0.23     |
| EWMA            | 0.346 | –2.96 (0.003)         | 36.3 | 0.64   | 0.18     |
| Yesterday's RV  | 0.466 | –5.38 ( $< 0.001$ )   | 35.0 | 0.58   | 0.33     |

□ 422 one-day-ahead forecasts of 24-hour realised variance; same models and tests as for SPY

## Reading the Bitcoin Results

- log-HAR wins again: QLIKE 22% below GARCH(1,1)- $t$  (DM  $t = -4.32$ ,  $p < 0.001$ )
  - ▶ GARCH on daily data is biased: MZ  $b = 0.55$  ( $p < 0.001$ ): it overreacts to large daily moves
- HAR in levels is worse than GARCH under QLIKE (0.409 vs 0.360): after calm weekends it predicts too low a variance, and QLIKE punishes under-prediction
- Intraday information can enter Bitcoin risk forecasts; entropy of high-frequency returns has been proposed as one such signal ([Pele & Mazurencu-Marinescu-Pele \(2019\)](#))
- Crypto-assets behave like a distinct asset class with their own volatility dynamics [Pele et al. \(2023\)](#)

## Forecasts over Time

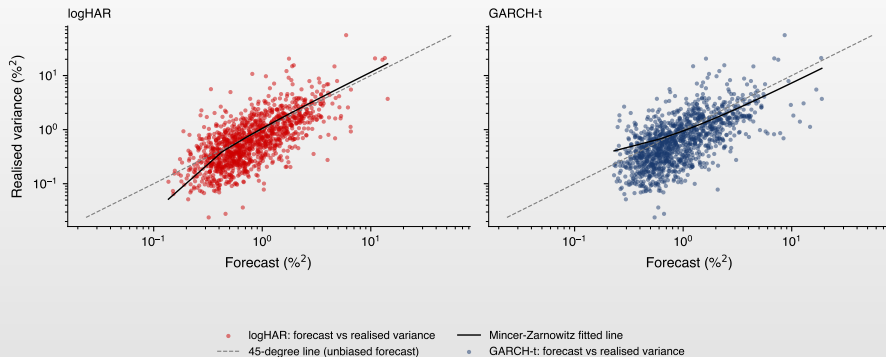


- Top: SPY realised volatility (with overnight) and the two forecasts, annualised; bottom: cumulative QLIKE loss of each model minus log-HAR (rising line = log-HAR better)

## Reading the Forecast Paths

- After a shock both forecasts jump; GARCH stays high longer (its persistence is close to 1), log-HAR returns faster
  - ▶ April 2025: the GARCH forecast stayed above 40% p.a. until late April, while realised volatility was below 20% from mid-April
- The cumulative loss difference rises steadily, with jumps at turbulent episodes: the gain is not driven by one day
- Of the three models plotted against log-HAR, EWMA loses most: a fixed  $\lambda = 0.94$  reacts too much to single days

## Mincer–Zarnowitz Regressions



□ SPY, 2022–2026: realised variance against the forecast (log scales); dashed: 45-degree line; black: fitted regression in levels

## Reading the Mincer–Zarnowitz Plots

- log-HAR forecasts span a wider range: they separate calm from turbulent days better
- Only 1% of GARCH forecasts are below  $0.26\%^2$  per day (log-HAR: minimum  $0.14\%^2$ ): on calm days realised variance falls far below GARCH
- The cloud is wide for both: one day's variance is hard to predict, and RV itself is measured with error
- Weekly horizon and model extensions: Seminar 9 (B6, B7)

## Beyond HAR: Semivariances and Joint Return–RV Models

- Realised semivariances  $RS_t^\pm = \sum_i r_{t,i}^2 \mathbf{1}\{r_{t,i} \gtrless 0\}$ ,  $RS^+ + RS^- = RV$ ; SHAR [Patton & Sheppard \(2015\)](#):  $\beta_d RV_t$  becomes  $\beta^+ RS_t^+ + \beta^- RS_t^-$ 
  - ▶ SPY, levels (overnight return counted as one more return of the day):  $\hat{\beta}^- = 0.56$  (0.23),  $\hat{\beta}^+ = 0.21$  (0.18); Wald  $\beta^+ = \beta^-$ :  $p = 0.35$
  - ▶ Bad volatility is more persistent, as in [Patton & Sheppard \(2015\)](#), but six years of data do not separate the two slopes
- Realized GARCH [Hansen, Huang & Shek \(2012\)](#): a GARCH equation driven by the realised measure  $x_t$  plus a **measurement equation**
  - ▶  $\ln h_t = \omega + \beta \ln h_{t-1} + \gamma \ln x_{t-1}$ ,  $\ln x_t = \xi + \varphi \ln h_t + \tau(z_t) + u_t$ ,  $r_t = \sqrt{h_t} z_t$ : joint quasi-maximum likelihood
- HEAVY models [Shephard & Sheppard \(2010\)](#): one GARCH-type equation for the daily variance, driven by lagged RV, and one for RV itself: consistent multi-step forecasts

## Which Models Survive? The Model Confidence Set

| Model           | QLIKE | MCS $p$ -value | In MCS (90%) |
|-----------------|-------|----------------|--------------|
| log-HAR         | 0.310 | 1.000          | ✓            |
| SHAR            | 0.320 | 0.224          | ✓            |
| HAR             | 0.326 | 0.014          |              |
| HARQ            | 0.344 | 0.095          |              |
| GARCH(1,1)- $t$ | 0.343 | 0.095          |              |
| EWMA            | 0.393 | 0.012          |              |
| Yesterday's RV  | 0.502 | 0.000          |              |

- Hansen, Lunde & Nason (2011): test equal predictive ability of all models (range statistic, block bootstrap), drop the worst, repeat; the surviving set contains the best model with probability  $\geq 90\%$
- SPY, 1 180 days from 2022: the set is {log-HAR, SHAR}; HARQ and SHAR are estimated in levels with the filter of Bollerslev, Patton & Quaadvlieg (2016) (forecasts outside the in-sample range replaced by the mean)
- A multiple-comparison procedure: many pairwise DM tests without size control would overstate the evidence (Chapter 8)

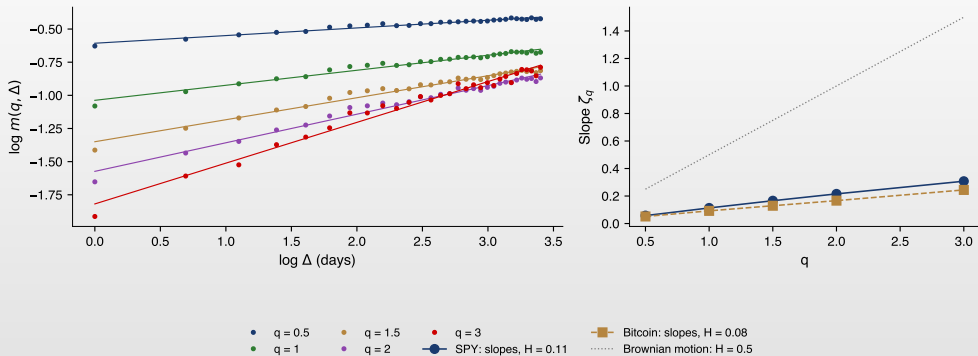
## Recap: Forecasting

- Evaluate variance forecasts with robust losses (QLIKE, MSE on variance) and test differences with Diebold–Mariano
- Intraday data help: log-HAR beats GARCH(1,1)- $t$  by 10% (SPY) and 22% (Bitcoin) in QLIKE
- Work in logs: HAR in levels loses under QLIKE when it under-predicts
- MSE rankings rest on a few crisis days and are rarely significant; the MCS keeps only log-HAR and SHAR for SPY

## Is Volatility Rough?

- Gatheral, Jaisson & Rosenbaum (2018): log volatility behaves like a **fractional Brownian motion** with Hurst exponent  $H \approx 0.1$ 
  - ▶  $H = 0.5$ : ordinary Brownian motion;  $H < 0.5$ : increments are negatively correlated, paths are rougher;  $H > 0.5$ : smoother, trending
- Scaling law:  $m(q, \Delta) = \frac{1}{T} \sum_t |\ln \sigma_{t+\Delta} - \ln \sigma_t|^q \propto \Delta^{qH}$ 
  - ▶ Regress  $\ln m(q, \Delta)$  on  $\ln \Delta$ : slope  $\zeta_q = qH$ ; then  $H$  from the slopes across  $q$
- Rough models reproduce the shape of the implied volatility smile for short maturities (Chapter 12)

## Scaling of Log Volatility Increments



□ Left:  $\ln m(q, \Delta)$  against  $\ln \Delta$  for SPY,  $\Delta = 1, \dots, 30$  days; right: slopes  $\zeta_q$  against  $q$

## Reading the Roughness Estimates

- Straight lines on the log-log scale: the scaling law holds; slopes are linear in  $q$ 
  - ▶  $\hat{H} = 0.11$  for SPY (block bootstrap 95% CI  $[0.09, 0.15]$ ), 0.08 for Bitcoin: far below 0.5
- Caution: RV measures volatility with error, and measurement error lowers  $\hat{H}$ 
  - ▶ A simulated Brownian motion gives  $\hat{H} = 0.46$ ; plus noise of a similar relative size,  $\hat{H} = 0.18$  (Seminar 9, B8)
- The scaling estimator on noisy RV proxies is **inconsistent** for the Hurst exponent; consistency needs high-frequency asymptotics [Fukasawa, Takabatake & Westphal \(2022\)](#)
  - ▶ Rough-looking estimates can come from estimation error in a non-rough model [Cont & Das \(2024\)](#); short-run roughness can coexist with long memory [Bennedsen, Lunde & Pakkanen \(2022\)](#)

## Recap: Rough Volatility

- The moments of log-volatility increments scale like  $\Delta^{qH}$  with  $H \approx 0.1$
- Our data:  $\hat{H} = 0.11$  (SPY) and 0.08 (Bitcoin)
- Part of the roughness can be measurement error in RV: i.i.d. error biases  $\hat{H}$  down, other error structures can bias it either way, so  $\hat{H}$  is not a bound on the latent  $H$
- Rough volatility models are an active research area in option pricing

## Case Study: Research Question

- Christensen, Siggaard & Veliyev (2023): does off-the-shelf machine learning (ML) forecast daily realised variance better than the HAR family, and if so, why?
  - ▶ HAR Corsi (2009) is linear in three RV lags, while RV is right-skewed and has long memory (stylised facts above): a nonlinear map may fit better
  - ▶ Its extensions (log-HAR, SHAR Patton & Sheppard (2015), HARQ Bollerslev, Patton & Quaedvlieg (2016)) still condition only on past returns
- Why other predictors rarely enter: OLS overfits when regressors are many, correlated and noisy
  - ▶ Candidates: implied volatility, the VIX, trading volume, macro and uncertainty indices (Table 1 of the paper)
- Why it matters: variance forecasts feed VaR (Chapter 7) and option pricing (Chapter 12); ML must beat a strong, three-parameter benchmark
  - ▶ Design choice: minimal hyperparameter tuning, so the test is of the methods “as is”, not of the tuning effort

## Case Study: Data and Method

- Data (Section 3): 29 of the 30 DJIA stocks (Visa excluded), 29 January 2001 – 31 December 2017,  $T = 4\,257$  days; 5-minute RV from cleaned NYSE trade prices,  $n = 78$  returns a day
  - ▶ Split: training 70% (2 964 days), validation 10% (424), test 20% (847); robustness: training 1 000 or 2 000 days, validation 200 (Appendix A.1)
  - ▶  $\mathcal{M}_{\text{HAR}}$ : daily, weekly, monthly RV;  $\mathcal{M}_{\text{ALL}}$  adds 9 predictors (Table 1): implied volatility, earnings-day dummy, 1-week momentum, change in log dollar volume; VIX, squared Hang Seng return, business-conditions index, change in the 3-month T-bill rate, policy-uncertainty index
- Models (Section 2): HAR, log-HAR, HAR with leverage, SHAR, HARQ by OLS on a rolling window; ridge ( $L_2$  penalty), lasso ( $L_1$ ), elastic net (a mix), adaptive and post-lasso, penalties tuned on the validation set
  - ▶ Trees: bagging, random forest (RF) with Breiman's defaults, 500 trees,  $J/3$  candidate predictors per split, minimum node size 5 (Table 15), gradient boosting
  - ▶ Neural networks  $\text{NN}_k^e$ :  $k = 1, \dots, 4$  hidden layers (2; 4-2; 8-4-2; 16-8-4-2 neurons), leaky ReLU, Adam, dropout, early stopping;  $e = 1$ : best of 100 seeds on validation,  $e = 10$ : average of the best 10
- Evaluation (Section 2.6): out-of-sample MSE relative to HAR, averaged over stocks; one-sided DM test per stock [Diebold & Mariano \(1995\)](#); MCS at 90% [Hansen, Lunde & Nason \(2011\)](#)

## Case Study: Key Results

- One day ahead, RV lags only (Table 2): MSE relative to HAR: log-HAR 0.995, HARQ 1.059, RF 1.020, neural networks 0.954–0.990
  - ▶ Past RV leaves little to learn: only the networks gain, by about 5%
- One day ahead, all 12 predictors (Table 3): OLS with all predictors 0.966, HARQ 1.289; elastic net 0.916, RF 0.901, log-HAR 0.901,  $NN_2^{10}$  0.885
  - ▶ Extra predictors pay only with shrinkage or nonlinearity; unpenalised HAR extensions overfit
- One month ahead, all predictors (Table 7): RF 0.604, bagging 0.621, networks 0.786–0.964, log-HAR 0.919, OLS with all predictors 1.290
  - ▶ The gain grows with the horizon: the ML fits are more persistent (Figure 8) and approximate the long memory of RV better
- MCS at 90% (Figure 4): with all predictors, the networks stay in the set for about 80% of stocks, HAR for about 20–40%
  - ▶ Accumulated local effects (Figure 7): daily RV, weekly RV and implied volatility dominate every model; RF and networks rank implied volatility higher than the linear models

## Case Study: Replicate, Limits, Extend

- Replicate on course data: SPY 5-minute bars (data/market/intraday), open-to-close RV with 78 returns a day, as in Section 3; the authors report that SPY confirms the stock results (footnote 12)
  - ▶ Split of Appendix A.1: 1 000 training days, 200 validation days, the rest to 18 September 2026 as test set
  - ▶ Predictors: VIX (it plays both the implied-volatility and the VIX role for an index fund), squared Hang Seng return, SPY momentum and dollar volume, T-bill and policy uncertainty (FRED); no earnings dummy for a fund (Seminar 9, C3)
- Limits stated by the authors
  - ▶ Minimal tuning: more tuning could improve ML further (Section 7); the networks are trained once and not re-estimated in the test window (Section 3)
  - ▶ Rankings depend on the split: with 1 000 training days the financial crisis falls in the test set and the HAR extensions beat HAR (Section 4.1, Appendix A.1)
  - ▶ HAR models with a separate jump component are not included (Section 2.2); accumulated local effects describe the average prediction and miss rare events such as earnings days (Section 4.2)
- Open extension: the paper ranks variance forecasts by MSE only; does the ML advantage survive QLIKE, which is robust to the noisy RV proxy [Patton \(2011\)](#), and does it hold for Bitcoin's 24-hour RV?

## An Open Question: Is Volatility Really Rough?

- Gatheral, Jaisson & Rosenbaum (2018) report  $H \approx 0.1$ , but  $H$  is estimated from RV, a noisy proxy of the latent  $\sigma_t$ 
  - ▶ Cont & Das (2024): the scaling estimator applied to realised volatility finds roughness even when spot volatility is smooth: an estimation artefact
  - ▶ Bolko, Christensen, Pakkanen & Veliyev (2023) (GMM that models the RV error) and Fukasawa, Takabatake & Westphal (2022) (consistent under high-frequency asymptotics) still find  $H < 0.5$
- Formal question:  $\ln \text{RV}_t^{1/2} = \ln \sigma_t + e_t$ , with  $\ln \sigma_t$  fractional of index  $H$ 
  - ▶  $H_0$ :  $H = 1/2$  (classical diffusive volatility) against  $H_1$ :  $H < 1/2$
  - ▶ Identification problem: a small  $H$  and a large  $\text{Var}(e_t)$  both flatten  $m(2, \Delta)$  at short lags
- Stakes: the short-maturity skew and volatility forecasts depend on  $H$  (Chapter 12)

## The Discovery Loop, Step by Step (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: tabulate the evidence, then check every DOI on [Crossref](#) (the title must match)
  - ▶ Tabulate the estimates of  $H$  published since 2018: estimator, sampling frequency, treatment of the RV measurement error, point estimate and standard error, DOI.
- 2. **Competing hypotheses**, each with its implied shape of  $m(2, \Delta)$ 
  - ▶ Derive  $m(2, \Delta)$  for three processes that give  $H$  near 0.1 with the scaling estimator: fractional log-volatility with  $H = 0.1$ ;  $H = 0.5$  plus i.i.d. error; a two-factor model with one fast mean-reverting factor.
- 3. **Data and code**: replicate the chapter's number first
  - ▶ Write Python that builds 5-minute RV from data/market/intraday/SPY.US\_5m.csv, computes  $m(q, \Delta)$  for  $q$  in  $\{0.5, 1, 1.5, 2, 3\}$ ,  $\Delta = 1..30$ , and the slope  $H$ . Target:  $\hat{H} = 0.11$

## The Discovery Loop, Step by Step (2/2)

- 4. **Robustness**, pre-registered, with a power check
  - ▶ Write a pre-registered plan: lag windows [1,30], [5,30], [10,50]; RV, bipower variation and 30-minute subsampled RV; a fit  $m(2, \Delta) = c \Delta^{2H} + k$ ; block-bootstrap intervals; size and power by simulation at  $H = 0.1$  and  $0.5$ .
- 5. **Critique**: ask for the strongest counter-argument
  - ▶ Act as a referee: which non-i.i.d. error structures in log RV (serially correlated, level-dependent, driven by overnight returns or jumps) would survive the constant- $k$  correction, and how would you detect each?
- 6. **Report**: all windows and measures, bootstrap intervals, the simulation design, code and the AI log

## What the Researcher Must Check

- **Proxy error:** RV is an estimate with a known standard error; test: simulate  $H = 0.5$  plus error of the same size and check that the procedure does not return  $H \approx 0.1$
- **Overlapping increments:**  $m(q, \Delta)$  reuses the same days at every  $\Delta$ , so OLS standard errors of  $\zeta_q$  are invalid; test: block bootstrap
- **Microstructure noise and jumps:** check the signature plot, then repeat with BV and with subsampled RV
- **Specification search:** lag windows and  $q$  grids chosen after seeing  $\hat{H}$  inflate significance; fix them in advance and report all
- **Invented estimates:** assistants quote " $H \approx 0.1$  for all assets"; check each number against the table of the cited paper

## Mini-Case: Is $\hat{H}$ an Artefact of i.i.d. Error?

- A typical assistant proposal:  $\hat{H} \approx 0.1$  comes from i.i.d. measurement error in RV. Implication:  $m(2, \Delta) = c \Delta^{2H} + k$  with  $k = 2 \text{Var}(e_t) > 0$ , so  $\hat{H}$  rises when short lags are dropped

|                                | $\Delta \in [1, 30]$ | [5, 30] | [10, 50] | Fit with $k$ , [1, 50] |
|--------------------------------|----------------------|---------|----------|------------------------|
| Simulated $H = 0.5$ plus error | 0.18                 | 0.24    | 0.26     | 0.39                   |
| SPY, total RV                  | 0.106                | 0.089   | 0.077    | 0.096                  |
| SPY, intraday RV               | 0.129                | 0.094   | 0.078    | 0.113                  |

- First column replicates the chapter; the simulated row (one path, not a power study) shows the mechanism:  $\hat{H}$  rises and the fit recovers most of the gap to 0.5
- SPY:  $\hat{H}$  falls with the window and the fitted  $k$  is 0;  $\hat{H} = 0.096$ , block bootstrap 95% CI [0.083, 0.203]
- Interpretation: an illustrative diagnostic, inconsistent with the i.i.d.-error explanation; a formal rejection needs a simulation-calibrated test; the non-i.i.d. mechanism of [Cont & Das \(2024\)](#) is not tested by this design. Code: `Quantlets/Ch_09/ai_discovery_case.py`

## Your Turn: A Project Seed

- Question: does the roughness of SPY and Bitcoin volatility survive an estimator that models the RV measurement error?
- Replicate first:  $\hat{H} = 0.11$  (SPY) and 0.08 (Bitcoin) from the scaling law, and  $\hat{H} = 0.18$  for simulated  $H = 0.5$  plus error (Seminar 9, B8)
- Deliverables
  - ▶ A pre-registration (hypotheses, estimators, lag windows, tests, simulation design) dated before the first run
  - ▶ The GMM estimator of [Bolko, Christensen, Pakkanen & Veliyev \(2023\)](#) on the course RV series, with Monte Carlo size and power at  $H \in \{0.1, 0.5\}$
  - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in AI\_USE.md and every error the assistant made in AI\_ERRORS.md: [github.com/danpele/MFM/tree/main/project](https://github.com/danpele/MFM/tree/main/project)

## Summary

- ▣ Realised variance turns intraday prices into a measurement of daily variance, with a known standard error
- ▣ Sample at 5 minutes for liquid assets; use noise-robust estimators when noise is material (tick data, illiquid assets); add the overnight return for daily risk
- ▣ Bipower variation, MedRV and truncation separate jumps from continuous variation; with FDR control jumps are rare and a small share of variance
- ▣ log RV is nearly Normal and has long memory: the log-HAR model captures it with three coefficients
- ▣ Out of sample, log-HAR beats GARCH for SPY and Bitcoin under QLIKE; MSE comparisons are inconclusive

## Which Tool When?

| Situation                              | Suggested tool                                                                     |
|----------------------------------------|------------------------------------------------------------------------------------|
| Liquid asset, 1–5 minute bars          | RV at 5 minutes, subsampled                                                        |
| Tick data, or an illiquid stock (BVB)  | Pre-averaging or realised kernel ( $n^{-1/4}$ ); check the signature plot          |
| Daily risk of an exchange-traded asset | Total variance: RV plus squared overnight return                                   |
| Separate news jumps from diffusion     | BV or MedRV, ratio test with FDR control; Lee–Mykland for the jump time            |
| Daily or weekly variance forecast      | log-HAR (plus jumps, weekends, RQ); benchmark GARCH                                |
| Comparing forecasts                    | QLIKE and MSE on variance, Diebold–Mariano, Mincer–Zarnowitz, model confidence set |

## Coming Next: Market Microstructure (Chapter 10)

- Where does the noise in high-frequency prices come from? The limit order book, bid–ask spreads, tick sizes
- Liquidity and market impact: how prices react to trading
- High-frequency trading and crypto exchange order books
- Further reading: Andersen, Bollerslev, Diebold & Labys (2003), Corsi (2009), Franke, Härdle & Hafner (2019)

## Further Study on Quantinar

### Advanced courses on quantinar.com

-  Nonparametric and Semiparametric Models
-  Kalman Filter
-  Applied Time Series Analysis with Python

### Code and data

- Quantlets: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_09](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_09)

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